Compiled Problems – Regular Expressions

1. **2010-11**

Which of the strings below are represented by the following regular expression?

$$01^*0\ (01^*10^* \cup 01^*00^*1 \cup 10^*1)\ 11^*001^*0$$

a) 0001101001 b) 0111001011 c) 010101010
d) 0100000 e) 011001011 f) 01010010

2. **2011-12**

Which of the following hexadecimal numbers, when written in binary, would be accepted by the following regular expression?

$$010^*\ (10^*\ (01^* \cup 01^*0) \cup 11^*0)$$

a) 48B b) 48E c) 5FE
d) 67E e) 577 f) 60E

3. **2015-16**

Given the following regular expression, simplify it using the identities to eliminate the union symbols.

$$a\ (a^* \cup b)^*\ b\ (b^* \cup a)^*$$

4. **Mine**

Which regular expressions are accepted by the regular expression: $a^*b\ (aab^* \cup baa^*)^*\ ba^*$?

A) abbbbabab B) baabaaaaa C) aaaaaaabbaaaaaaaaaaa
D) baabbbbaab E) bbbabbb F) abababaaba

5. **Mine**

Which regular expressions are accepted by the following regular expression (u=A,…,Z and l=a,…,z)?

$$ull^*_u._ul^*$$

A) Ullyses_S._Grant B) Ms_ C._Blackstone C) Emmaus_H._School
D) U_S._A E) Abe_Lincoln F) Madam_Im_Adam

6. **Mine**

Which regular expressions are equivalent to the following regular expression?

$$ab^*\ (ba^* \cup ab^*)\ ba^*$$

A) $ab^*ba^*ab^*ba^*$ B) $ab\ (b^*a^* \cup b^*ab)\ b^*a^*$ C) $ab^*ba^*ba^* \cup ab^*ab^*ba^*$
D) $(ab^*ba^*ba^*)^*$ E) $ab^*\ (ab^* \cup ba^*)\ ba^*$ F) $abb^*a^*ba^* \cup ab^*abb^*a^*$

7. **Mine**

Which regular expressions are accepted by the following regular expression (d=0,…,9)?

($ ∪ d*) d. (dd ∪ λ)

A) $0.45 B) $32.90 C) $8.456
D) 435.7 E) 5.43 F) $6.

8. **Mine**

Which regular expressions are equivalent to the following regular expression?

(ab*ba*)*

A) a*b*a* B) (a*b*)* C) ab*ba*
D) (ab*)*(ba*)* E) (ab)* F) a*b*a*

9. **Mine**

Which regular expressions are accepted by the following regular expression (f=a,…,m and s=n,…,z)?

ff*s*s ∪ (sf)*

A) fffffssssss B) fsfsfsfs C) sfsfsfsf
D) λ E) sssssssffffff F) fs

10. **Mine**

Which regular expressions are accepted by the following regular expression?

1 (1 ∪ 0)* 0*0

A) all even numbers B) all binary numbers C) all starting with 1 and ending with 0
D) all odd numbers E) all multiples of 4 F) all with alternating 0s and 1s